

Security Engineering

ECE 371

Taqi Raza, Fall 2025

Meeting 1

Who is Taqi Raza?

taqi@umass.edu

- Received PhD and MS from UCLA in Computer Science
- Major research area is on network + systems security
- Assistant professor at UMass Amherst ... this is my third year!
 - Previously, worked as an assistant professor at the University of Arizona for 3 years.
- Had 5 years work experience in South Korea (LG Electronics)

Labs are Led by

- Prof. Nikhil Saxena (nikhilsaxena@umass.edu)
 - Marcus Hall, Room 213

371 Staff

- Head TA: Muhammad Shayan Nazeer <mnazeer@umass.edu>;
 - UIA Lab : Khushii Shetty <khushiishett@umass.edu>;
 - UIA Lab : Anika Badkul <abadkul@umass.edu>;
 - UIA Lab : TBD
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- Technician: Luke LaCasse <llacasse@umass.edu>;

Attendance

- You are expected to attend all lectures physically in the class.
- We do activities in class that need your participation.
- We will use random Quizzes to take attendance and check on how attentive you are in the class.
 - I will take off bottom 3 quizzes from your grading.
- Quizzes have NO Make-ups

NO Excuse for missing the Class

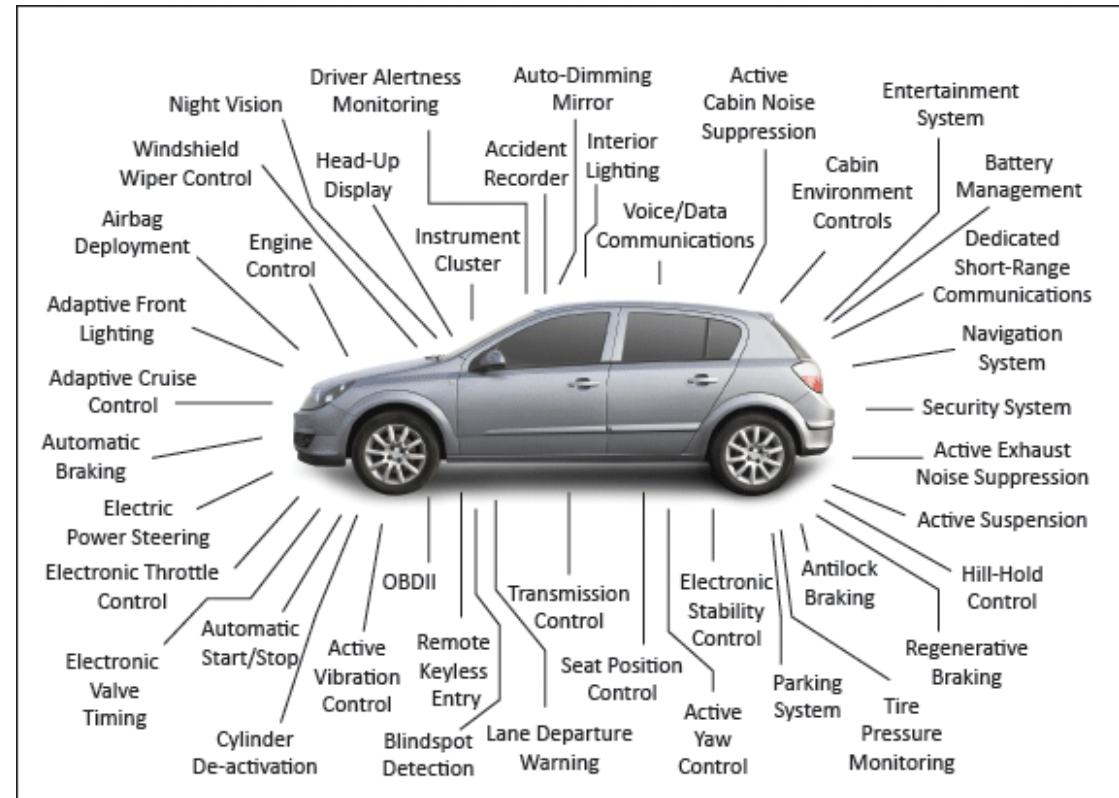
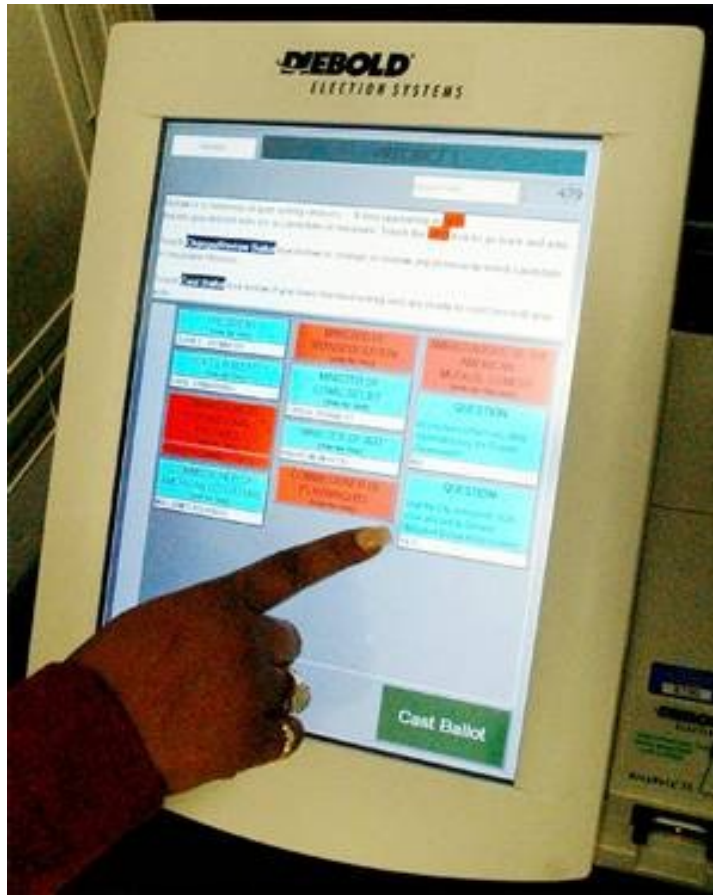
- Only family/personal emergencies, leave for mandatory duties are accepted
- Other excuses are NOT accepted; examples
 - I am overwhelmed with other courses
 - My answer: Should have done the time management
 - I have an interview at the same time
 - My answer: There are other dates and times in the week
 - I have to attend the job fair
 - My answer: 50 minutes class can be taken off from on-campus job fair; you may have come a bit late too.
 - I have to fly home early for ThanksGiving
 - My answer: UMass is not closed

Quizzes have NO Make-ups. Either you take it or not!

What is Security Engineering?

- A multi-disciplinary field of engineering focusing on security aspects of system design
- Focus on making sure the system is dependable when attacked by adversaries, malware
- Tools, procedures, design methodologies, test methods,
- Cryptography, computer hardware, systems engineering, economics, statistics, psychology,

Examples of Secure Systems



Examples of Secure Systems (cont)



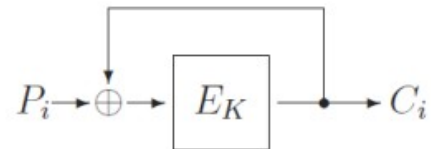
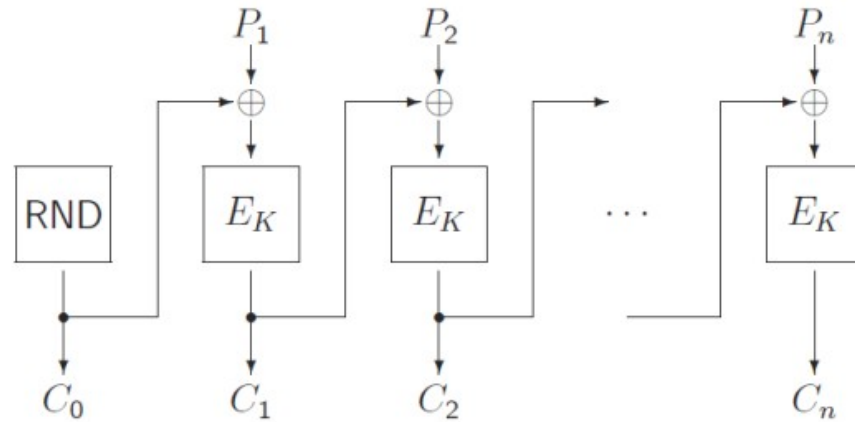
Transportation Payment System



Medical Devices

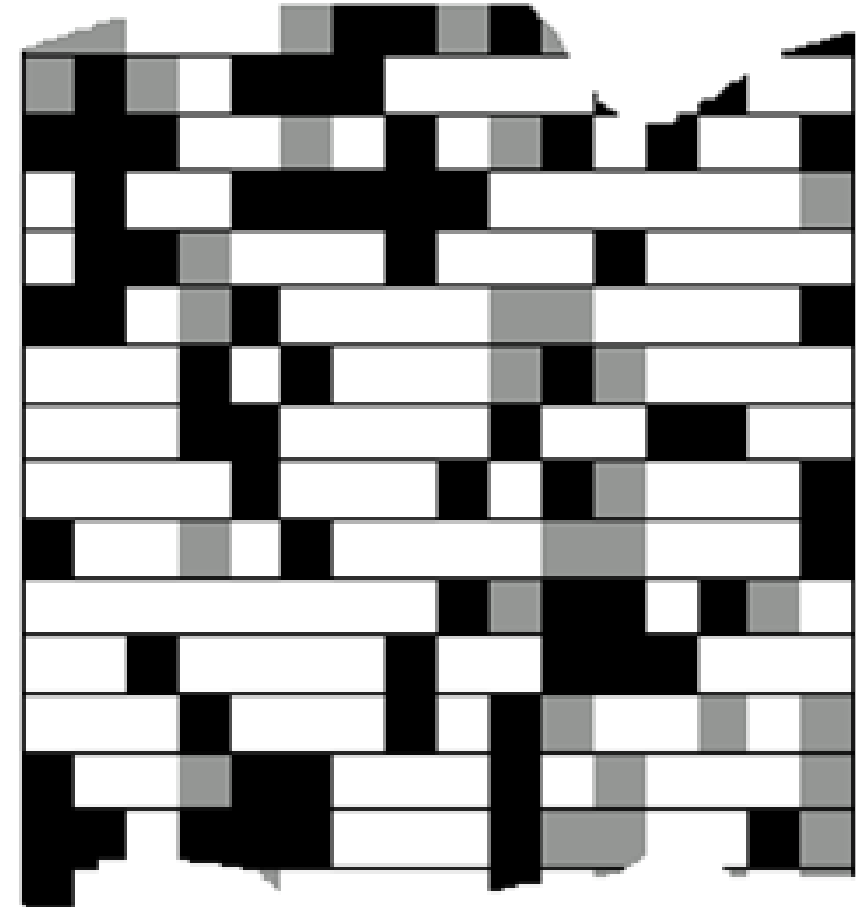
Representations of Secure Systems

Cipher Block Chaining (CBC)



$$C_i = E_K(P_i \oplus C_{i-1})$$

Algorithms



Information and Statistics

Course Objectives

- Examine the foundations, challenges, and trends in security engineering
- Apply guidelines for secure system design
- Distinguish between recent vulnerabilities and defenses in security engineering
- Classify examples of hardware security issues (EE)
- Identify the importance of non-technical issues within the field of security engineering
- Discover futuristic applications of security engineering principles

Definition of System Security

“The science of managing malicious intent and behavior that involves information and communication technology.”

Security vs. safety engineering: focus on **intentional** rather than **accidental** behavior, presence of **intelligent adversary**.

Definition of System Security (cont'd)

- Malicious behavior can include:
 - Fraud/theft - unauthorized access to money, goods or services
 - Vandalism - causing damage for personal reasons
 - Terrorism - causing damage, disruption and fear to intimidate
 - Warfare - damaging military assets to overthrow a government
 - Espionage - stealing information to gain competitive advantage
 - Sabotage - causing damage to gain competitive advantage
 - “Spam” - unsolicited marketing wasting time/resources
 - Illegal content - illicit and/or extremist materials

Trends in System Security

- Applications
 - Digital wallet and keys (banking, payments, medical, voting, access)
 - Business models based on security/privacy (social networking, games)
 - Data-centric computing (images, audio, video, graphics, 3D,...)
- Platforms
 - Mobile clients decreasing cost, size, power
 - Demand for increased reliability and usability
 - Constant software updates
 - Virtualization
 - Ubiquitous connectivity to cloud

Trends in System Security (cont'd)

- Hardware
 - Multi-core, heterogeneous, accelerators, configurable
 - More on-chip cache
 - More off-chip memory, increasingly non-volatile
 - Removable storage (USB, etc.)
 - Increasing Wireless Connectivity (WiFi, 5G, GPS, BT, NFC,...)
- Threats
 - Increasing global cyber-crime at all levels
 - “Attackers” include: Nation-state, terror, organized crime, petty crime, hackers, user!
 - Increasing privacy concerns for private citizens and companies

How are Systems Hacked?

- **Hack attack:** low-cost remote software attack
 - User often inadvertently approves the installation of the software that then executes the attack
 - Examples: viruses and malware downloaded to the device via a physical or a wireless connection
- **Shack attack:** low-cost hardware attack
 - Attacker uses equipment from retail electronics store with designs and components found on the Internet.
 - Example: electronic IED
 - Attackers have physical access to the device, but not enough equipment or expertise to attack within the integrated circuit packages

(ARM Limited, 2009)

How are Systems Hacked? (cont'd)

- **Lab attack:** the most comprehensive and invasive
 - Attacker can use laboratory equipment to perform reverse engineering of sensitive device parts
 - Example: Attach microscopic logic probes to silicon metal layers, and glitch a running circuit using lasers
 - Example: Monitor analog signals, such as device power usage and electromagnetic emissions, to perform cryptographic key analysis
- Rule of thumb: Every device can be broken
 - You do not defend against attack directly, but limit the damage caused

(ARM Limited, 2009)

The Rise and Risk of Embedded Devices

- **Definition:** Highly specialized device meant for specific purpose(s), usually embedded within another object or as part of a larger system
 - Example: Heart rate monitor embedded in wristwatch that connects to smartphone
- **Forecast:** 100 billion embedded devices in use now with exponential trend
 - Connected to all aspects of life, from critical infrastructure to daily activities
- **Warning:** Embedded devices have now become the targets of criminals
 - Example: Attacks on industrial control systems

How to Talk about Security?

The subject of security engineering covers both problems and solutions, but that information is not always used by those with good intentions.

“In previous centuries, people objected to the publication of books on lock-smithing, on the grounds that they were likely to help the bad guys more than the good guys.”
Anderson, 2008

Should the topic of security “hacks” be taught without restriction? Why or why not?

What is Security Engineering?

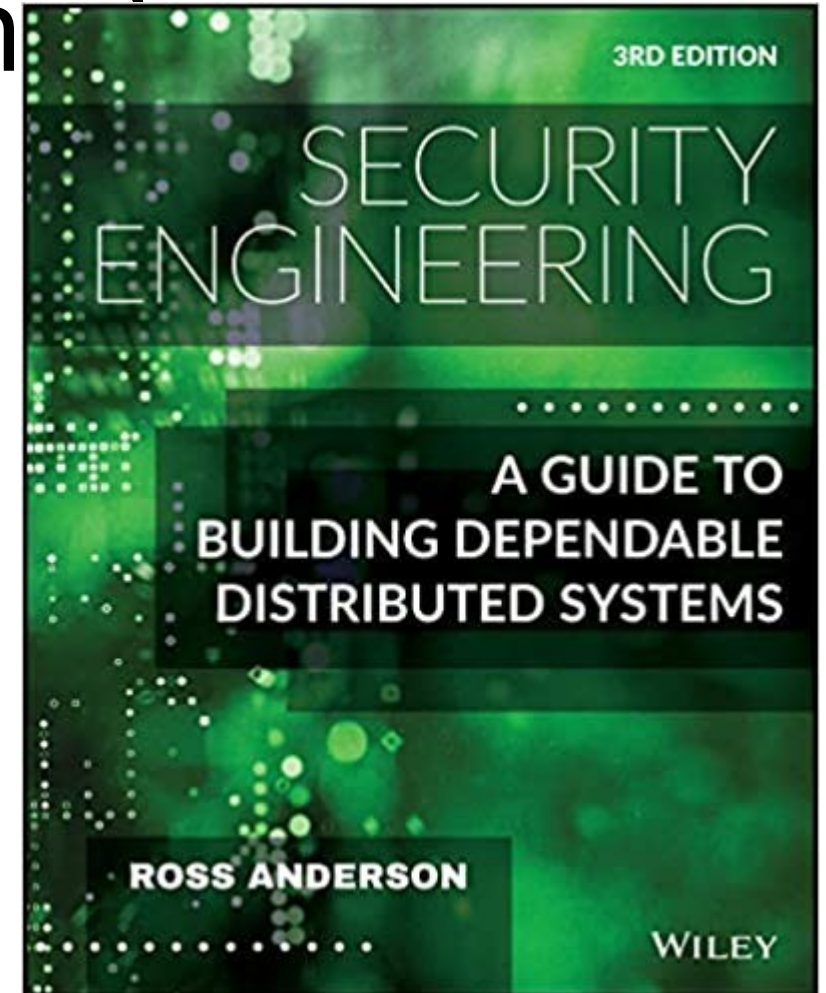
- Course textbook:

Anderson, R. (2021). *Security engineering: A guide to building dependable distributed systems* (3rd edition).

Wiley. Available in digital edition from Amazon or Wiley for about \$50. Used copies may be available.

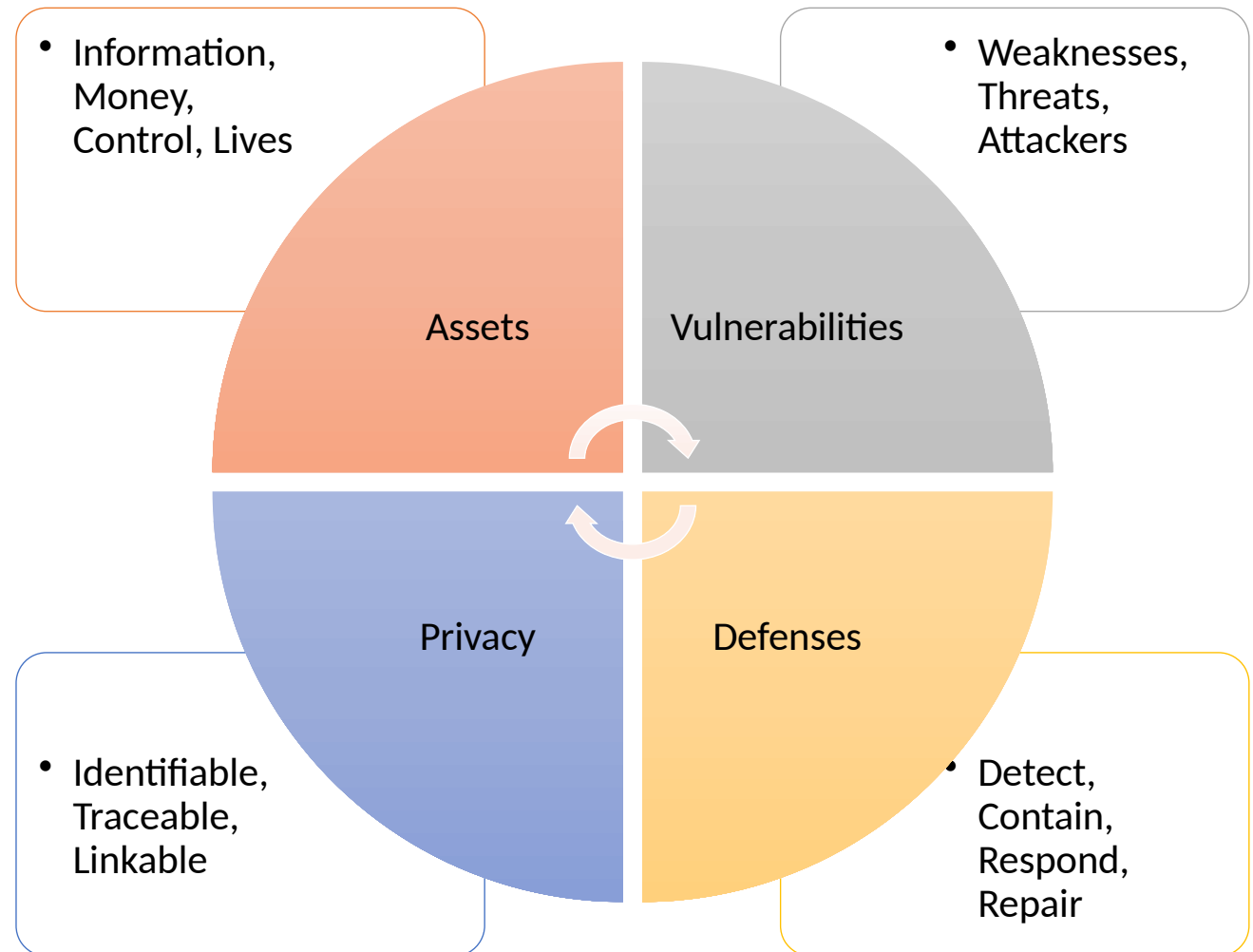
Digital Copy: <https://www.cl.cam.ac.uk/archive/rja14/book.html>

“Security is not just about keeping the bad guys out, but increasingly concerned with tussles for power and control.”



What is a Secure System?

- Assets (Information, Money, Control, Lives,...)
- Vulnerabilities (Weaknesses, Threats, Attackers)
- Defenses (Detect, Contain, Respond, Repair)
- Privacy (Identifiable, Traceable, Linkable,...)



Small Group Discussions

- Some kind of opinion question, probably not technical.
- 5-10 minutes
- Everyone speaks at least once
- One person will report back (consider some notes)
- I will drop in randomly to see how it is going.

Small Group Discussion: Additional Secure System Examples

- Bank
- Hospital
- Home
- Industrial Control System
- Public transit station

For each, think about

1. Assets (Information, Money, Control, Lives,...)
2. Vulnerabilities (Weaknesses, Threats, Attackers)
3. Defenses (Detect, Contain, Respond, Repair)
4. Privacy (Identifiable, Traceable, Linkable,...)

A Defense Framework

- Examples:
 - Airport Security
 - Olympics 2020

Policy

*What to
do?*

Mechanism

*How does
it work?*

Assurance

*How to
make sure
it works?*

Incentives

*How do
you
encourage
it?*

Why is Information Security Hard?

- Common view:
 - Security is a difficult technical issue
 - Cryptography, tamper-proofing, law enforcement
- But other factors also make things difficult
 - Business disincentives
 - Human behavior

Security Economics

- **Remember: With enough resources, every device can be broken**
- Need to consider economics of attack and defense

Security Economics (cont'd)

- What is the **cost** of defense in terms of implementation and loss of utility?
- What is the **value** of stored assets?
 - Application-specific (payment, appliance, medical, automotive, etc.)
 - What data is being sought?
 - What is the value of the data (\$\$, privacy)
 - Can the data be used for further attack (e.g. keys, passwords, CC #s)
 - Other objectives besides data (e.g. control, DOS, cloning)

Security Economics (cont'd)

- Therefore, limit the damage when a device is broken and therefore make the lab attack **uneconomical**.
 - Use of *per-device unique secrets* is one example where reverse engineering a single device provides the attacker with no useful information

Incentives in Security Economics

- Who has primary responsibility when bank fraud occurs?
 - In US: the bank
 - In UK: the customer
- Guess who has the more effective security system?

Hot Topics In the News (a regular feature)

- Digital Wallets
- Microsoft Azure
- Machine learning for attack and defense
- How to get involved:
 - Hacker-one, Google Bug Bounty, Esports for Cyber
- Ransomware
- Banks
- Venmo
- Cisco VM bug
- Trust but Verify

Syllabus

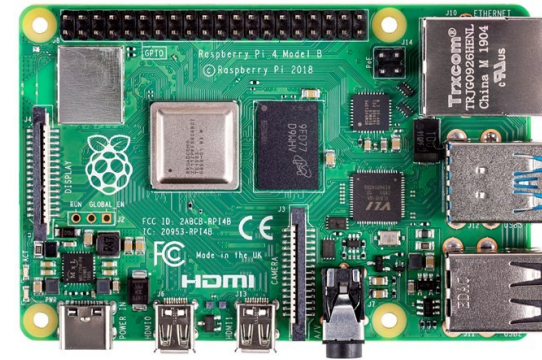
- Logistics
 - Canvas
 - TA & UIA Office Hours: TBD
 - Lab partners (up to 2 students per group for the lab assignment)
- Grading
 - Lab demos and reports
 - Exams: Midterm and Final
 - Homeworks
 - Quizzes
- Book and Readings (all on Canvas)
- Questionnaire (please turn in asap!)

Reading Assignment

- Introduction, Preface and Chap 1 in Anderson textbook (2021, 3rd edition)
[Security Engineering - A Guide to Building Dependable Distributed Systems](#)
)
- Anderson, R. (2001, December). “Why information security is hard: An economic perspective.” Find this on IEEE Xplore.

Labs (40% of the total grade)

- Find a Lab partner asap!
- Everyone gets their own Raspberry Pi, and the card-key (to be activated by students) to Duda Hall.
- 5 Labs
 - Lab 1 - Buffer Overflow
 - Lab 2 - AES Encryption of Images
 - Lab 3 - Network Security
 - Lab 4 - Formal Methods/Fuzzing
 - Lab 5 - TBD
- Each Lab will be graded based on a demo and a report.



Duda Hall, starting next week

- One sign up form per group. 2 students per group.
- No groups of 1 or 3 allowed without Prof. Taqi's approval.
- Everyone read and sign the form.
 - Return the form to Technician, Luke LaCasse.
- Luke will give you the Duda Hall access key cards
 - You need to activate the key card.
 - Luke will give you the lab kit.
- If any of your group members are not present, they will have to e-mail Luke <llacasse@umass.edu> to make an appointment for the key card and the kit.

Labs

- Start working on labs as soon as they are assigned
- Starting lab late is not an excuse and will not be accepted
 - Working only close to the deadlines will not give you enough time for debugging issues (and cater for other personal time off)

Quizzes (10% of the total grade)

- Randomly, about 1/week.
- During start, middle or the end of lecture, covering material from that lecture
- Use laptop or phone to access Canvas quiz.
- Contact me for any questions
- Objective: come to lectures and verify basic concepts and skills.

Exams and Assignments

- Assignments/homeworks – 10% of the total grade
- Midterm – 20% of the total grade
- Final Exam – 20% of the total grade
- Class survey (extra credit) – 0.5% of the total grade



College of Engineering Student Organizations & Resource Fair

Tuesday September 9, 2025, 10:30 AM - Tuesday September 9, 2025, 3:30 PM

Engineering Quad

[Community](#)

On September 9 from 10:30 a.m. to 3:30 p.m., the College of Engineering (CoE) will hold its Student Organization and Resource Fair in a large tent set up on the CoE quad, in front of Marston Hall. The event will inform new and returning CoE students about our numerous student organizations and the college's support resources, including the **Career Development and Experiential Learning Center**, the **Office of Student**